



Heaven's light is our Guide

Department of Electrical & Electronic Engineering

Rajshahi University of Engineering & Technology

Design Project on Electrical Service Design (EEE 4100)

Marks		Course Code	EEE 4100
Allocated	Obtained	Course Title	Electrical Service Design
20		Semester & Year	4 th Year Odd Semester (20 series)
		Course Instructor	Prof. Dr. G. K. M. Hasanuzzaman
			Alok Kumar Paul
			Sakib Hossen
		Due Date	
		Submission Date	6-1-2026
Student ID	2001074	Student Name	Shahziar Karim Safit
Submission Deadline		06 January 2026	

General Guideline:

Printed Document:

Print this document and attach your design project at the end of this file before submission.

Viva Voce (If required):

A viva (oral examination) may be conducted to verify or clarify your submitted design project.

Problem statement [CO1-PO(e), CO1-PO(f), CO1-PO(h)]:

Each student must prepare a **complete electrical service design** for a modern apartment building that includes the following components:

1. Conduit Layout

- ✚ Develop detailed conduit routing plans showing the placement, type, and size of conduits.
- ✚ Include power, lighting, and communication circuits.
- ✚ Ensure compliance with BNBC 2020 requirements for installation and accessibility.

2. Fixture Layout

- ✚ Prepare detailed layouts for lighting, ceiling fans, and power outlets.
- ✚ Account for dedicated circuits for heavy-load appliances (e.g., Air conditioner, electric cooking, washer/dryer).

3. Communication Layout

- ✚ Design structured cabling for telephone, internet, intercom, and television systems.
- ✚ Maintain proper separation between communication and power circuits to prevent interference.
- ✚ Indicate communication panel locations and connection points.

4. Lightning Protection System

- ✚ Design a complete lightning protection and grounding network.
- ✚ Include air terminals, down conductors, and earthing arrangements with proper resistance values.
- ✚ Ensure adherence to BNBC and international safety standards.

5. 11kV Substation (Single Line Diagram)

- ✚ Prepare a **Single Line Diagram (SLD)** showing the 11kV substation layout.
- ✚ Include transformer specifications, main switchgear, circuit breakers, and protective devices.

- ✚ Clearly indicate ratings, cable interconnections, and metering arrangements.

6. BOQ/BOM (Bill of Quantities / Bill of Materials)

- ✚ Provide a complete list of materials and components used in the design.
- ✚ Include item descriptions, quantities, specifications, unit prices, and total cost.

Design Requirements

- Select appropriate **cable types, sizes, and protective devices** (MCBs, RCCBs, etc.) based on calculated loads.
- Ensure the total connected load **does not exceed the utility's permitted capacity.**
- Incorporate **energy-efficient and low-power equipment** to promote sustainability.
- Ensure **compliance with safety, environmental, and health standards.**
- Maintain balance between **performance, cost, and sustainability.**

Uniqueness Rules

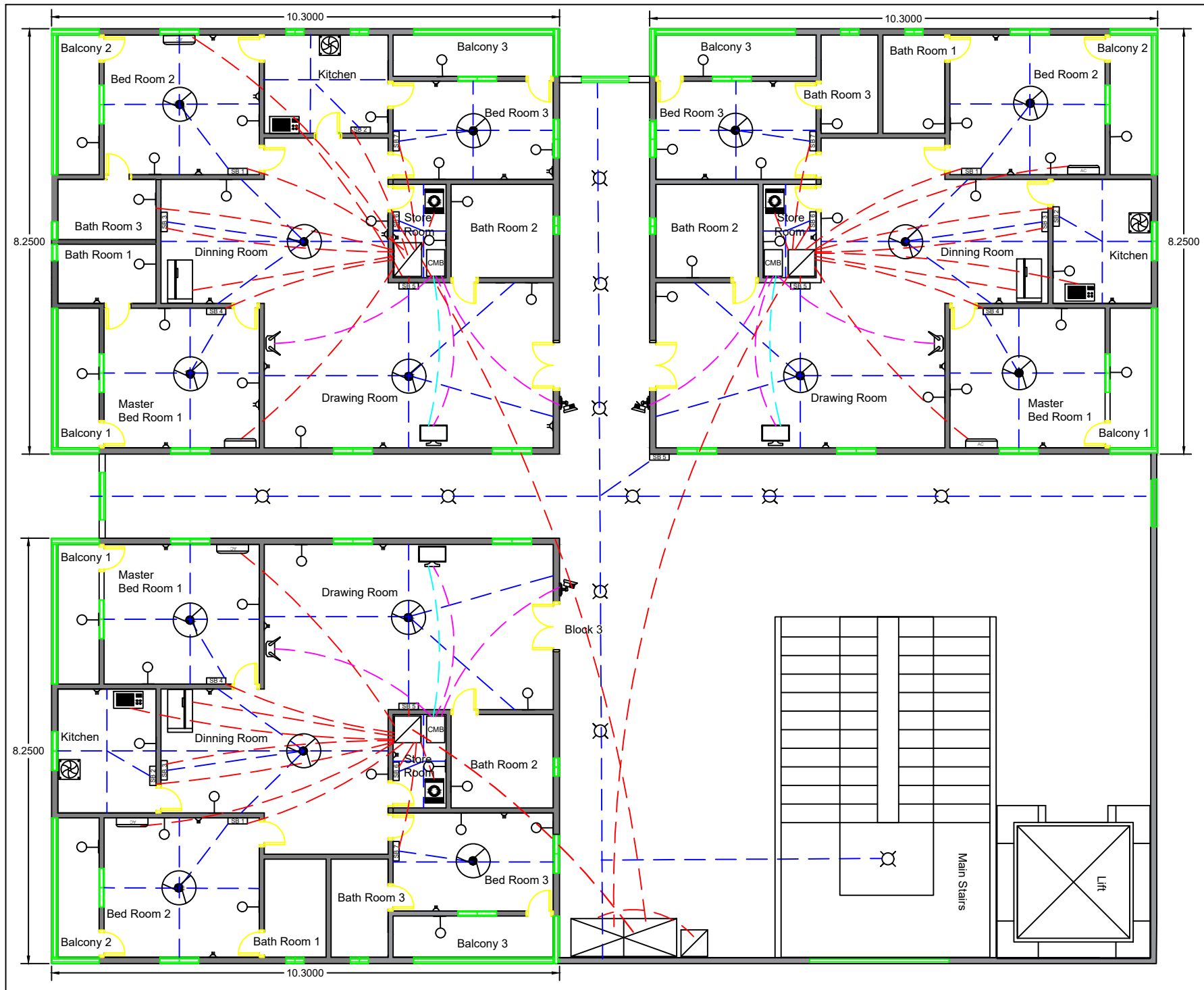
Last Roll Digit	No. of Floors	Apartments / Floor	Typical Apartment Size	Minimum % of the Air conditioner of the total load
0	8	3	95 m ²	20
1	7	4	90 m ²	25
2	9	3	100 m ²	30
3	10	4	105 m ²	20
4	6	3	85 m ²	25
5	8	4	92 m ²	30
6	9	3	110 m ²	20
7	7	4	88 m ²	25
8	10	3	120 m ²	30
9	6	4	80 m ²	20

** Finally, justify how the corresponding COs and POs are attained through this assignment.

Marking Rubrics:

Criteria (Mapped to POs)	Excellent	Good	Average	Needs Improvement
PO(e): Design, Analysis & Tool Usage (10 Marks)	All layouts (Conduit, Fixture, Communication, Lightning) and the 11kV SLD are technically sound, detailed, accurate, and clearly presented. Load calculations and selection of all components (cables, protective devices) are correct and explicitly justified. (9-10 Marks)	Layouts and SLD are mostly complete and correct but contain minor technical errors, omissions, or lack clarity. Load calculations or component selections have minor errors but are generally appropriate. (7-8 Marks)	Layouts or SLD are incomplete or contain significant technical errors. Load calculations are flawed or key components (e.g., transformer, main switchgear) are incorrectly specified. (5-6 Marks)	Major design components are missing or fundamentally flawed. Layouts are incorrect, and load calculations are absent or fail to meet basic requirements. (3-4 Marks)
PO(f): The Engineer & Society (Safety & Standards) (5 Marks)	The design explicitly and correctly adheres to all required standards, including BNBC 2020, international safety standards, and utility capacity limits. Safety, environmental, and health considerations are clearly integrated. (5 Marks)	The design generally follows standards but has minor deviations or omissions (e.g., minor accessibility issues, incorrect circuit separation). Safety considerations are mentioned but not fully integrated. (4 Marks)	The design overlooks key standards (e.g., BNBC requirements, grounding) or safety factors. Compliance is asserted but not demonstrated in the design. (3 Marks)	The design shows a clear disregard for, or lack of awareness of, fundamental safety standards and legal requirements (BNBC, utility rules). (2 Marks)

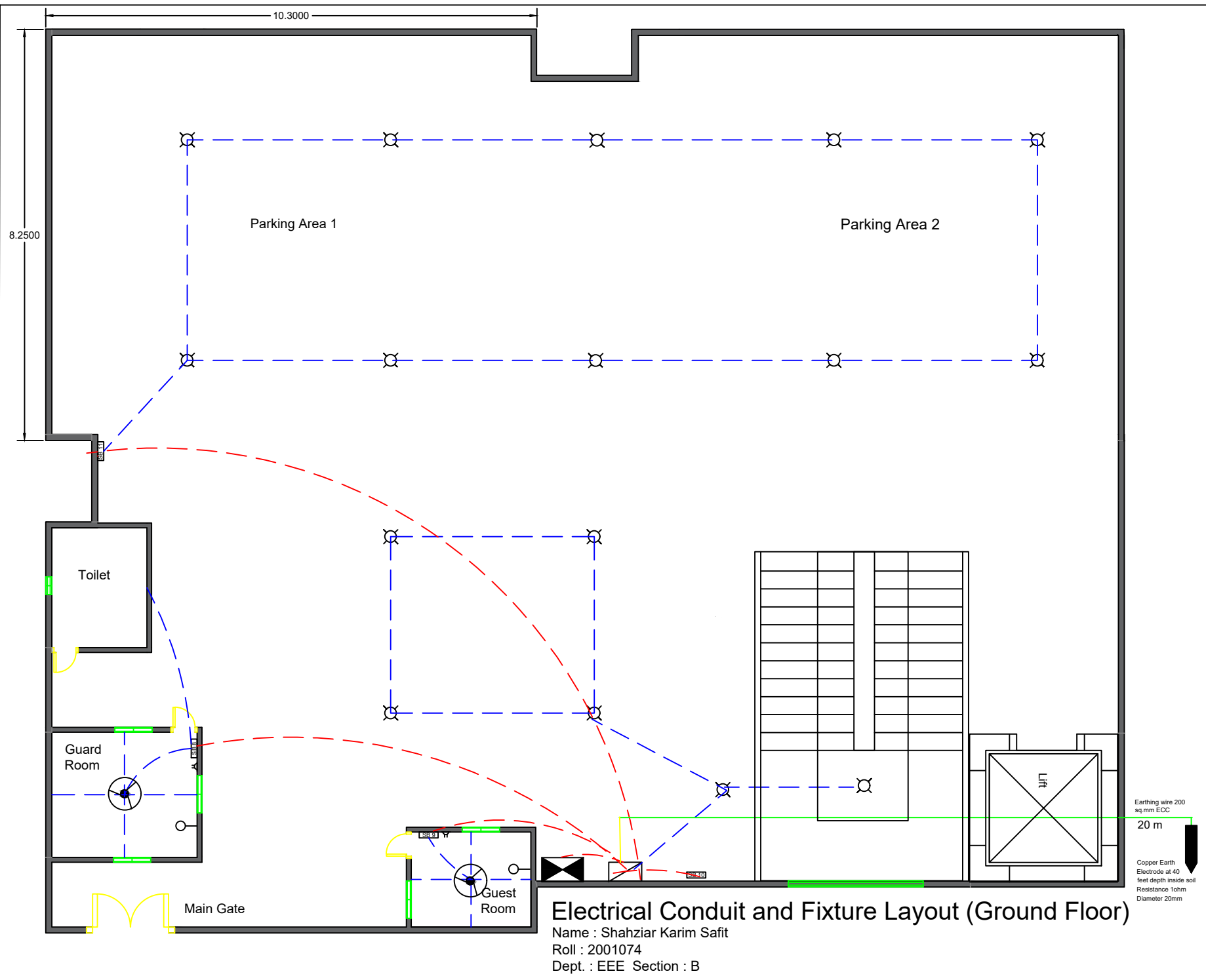
PO(h): Ethics, Sustainability & Professionalism (5 Marks)	BOQ/BOM is comprehensive, professional, and accurate, with detailed items, specs, and costs. Sustainability is effectively promoted via energy-efficient equipment, and the balance between cost, performance, and sustainability is well-justified. (5 Marks)	BOQ/BOM is mostly complete but has minor omissions or costing errors. Sustainability is addressed, but the cost/performance balance is not well-justified. (4 Marks)	BOQ/BOM is incomplete, with missing items or specs. Sustainability or energy efficiency is not adequately addressed. (3 Marks)	BOQ/BOM is missing or deeply flawed. There is no consideration for sustainability, cost, or professional ethics. (2 Marks)
Late Submission Penalty	Submitted on time: No Penalty 1 day late: -1 mark > 1 day late: -3 marks			

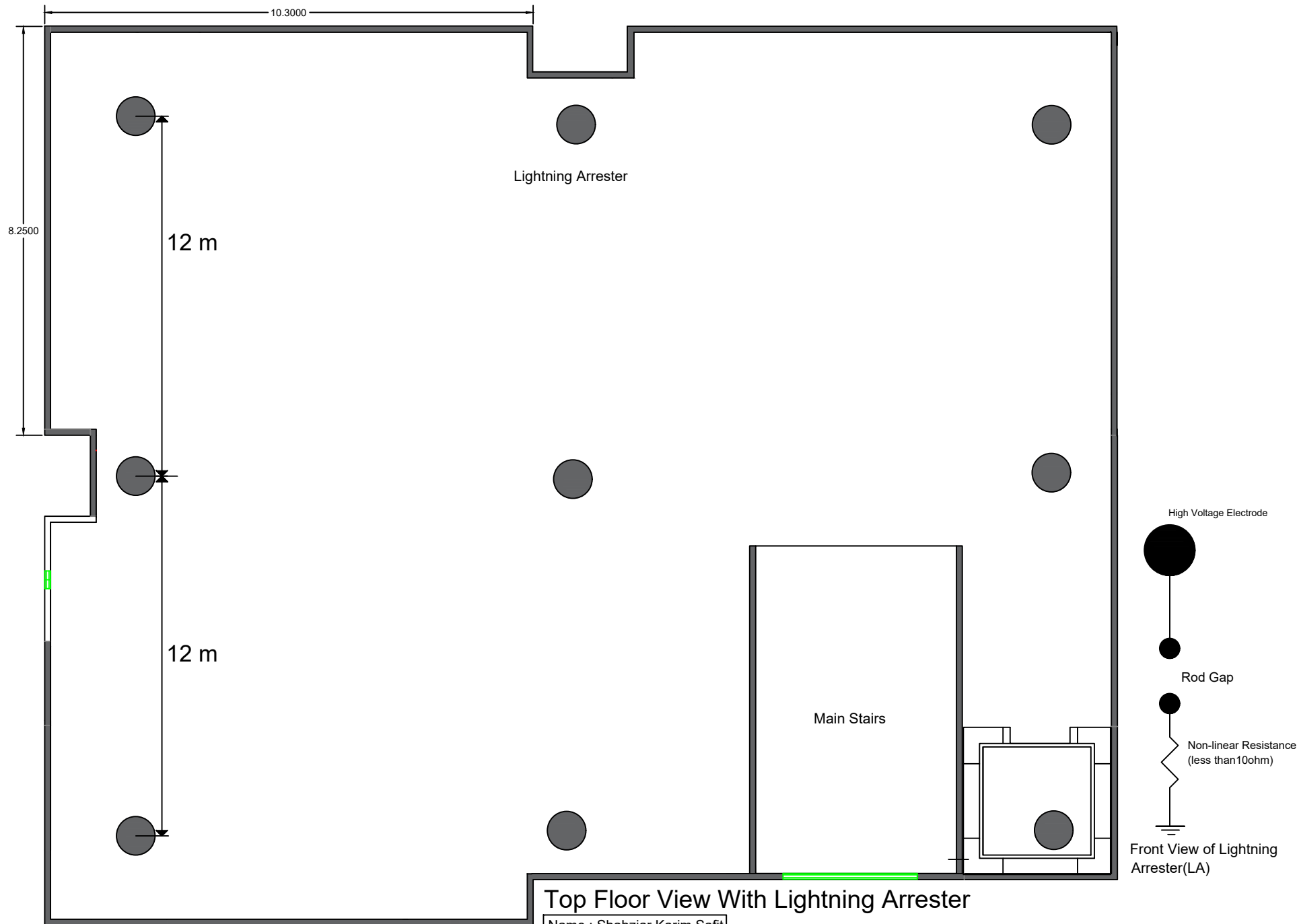


LEGEND	
SYMBOL	DESCRIPTION
	Ceiling Fan
	Wall Light
	Tube Light
	3 Pin Socket
	2 Pin Socket
	Exhaust Fan
	AC
	Oven
	Refrigerator
	Television
	CC Camera
	Router
	Washing Machine
	Communication Board
	Switch Board
	Sub Distribution Board (SDB)
	Floor Distribution Board (FDB)
	Main Distribution Board (MDB)
	Lightning Arrester
	Conduit
	Power Load Conduit
	Communication Cable
	Dish Cable

Electrical Conduit & Fixture Layout
(Middle Floor)

Name : Shahziar Karim Safit
Roll : 2001074
Dept. : EEE Section : B





Top Floor View With Lightning Arrester

Name : Shahziar Karim Safit
Roll : 2001074
Dept. : EEE Section : B

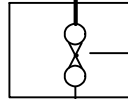
MCB Layout for Each Floor (2-6)

Name : Shahziar Karim Safit

Roll : 2001074 Dept: EEE Section: B

MDB

E



Fuse

FDB

LEGEND



MCB



Power Load wire



Light Load wire



Switch Box

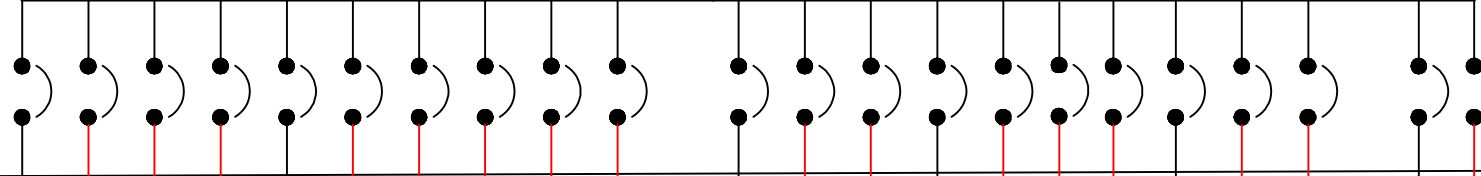
SDB 1



DP MCB

MCB

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21



SDB 2

SDB 3

1 x 2.5 mm² BYA

1 x 1.5 mm² BYA

SB1
WL1
WL2
WL3
2pin
CF

3 Pin

3pin

AC

SB2
WL4

3 Pin

Oven

SB3

3 Pin

Refrigerator

SB4
WL9
WL10
WL11
2PIN
CF

3 Pin

AC

SB5
WL12
WL13
CF
2PIN

3 Pin

3 Pin

TV

SB6
WL14
WL15
2 PIN
2PIN

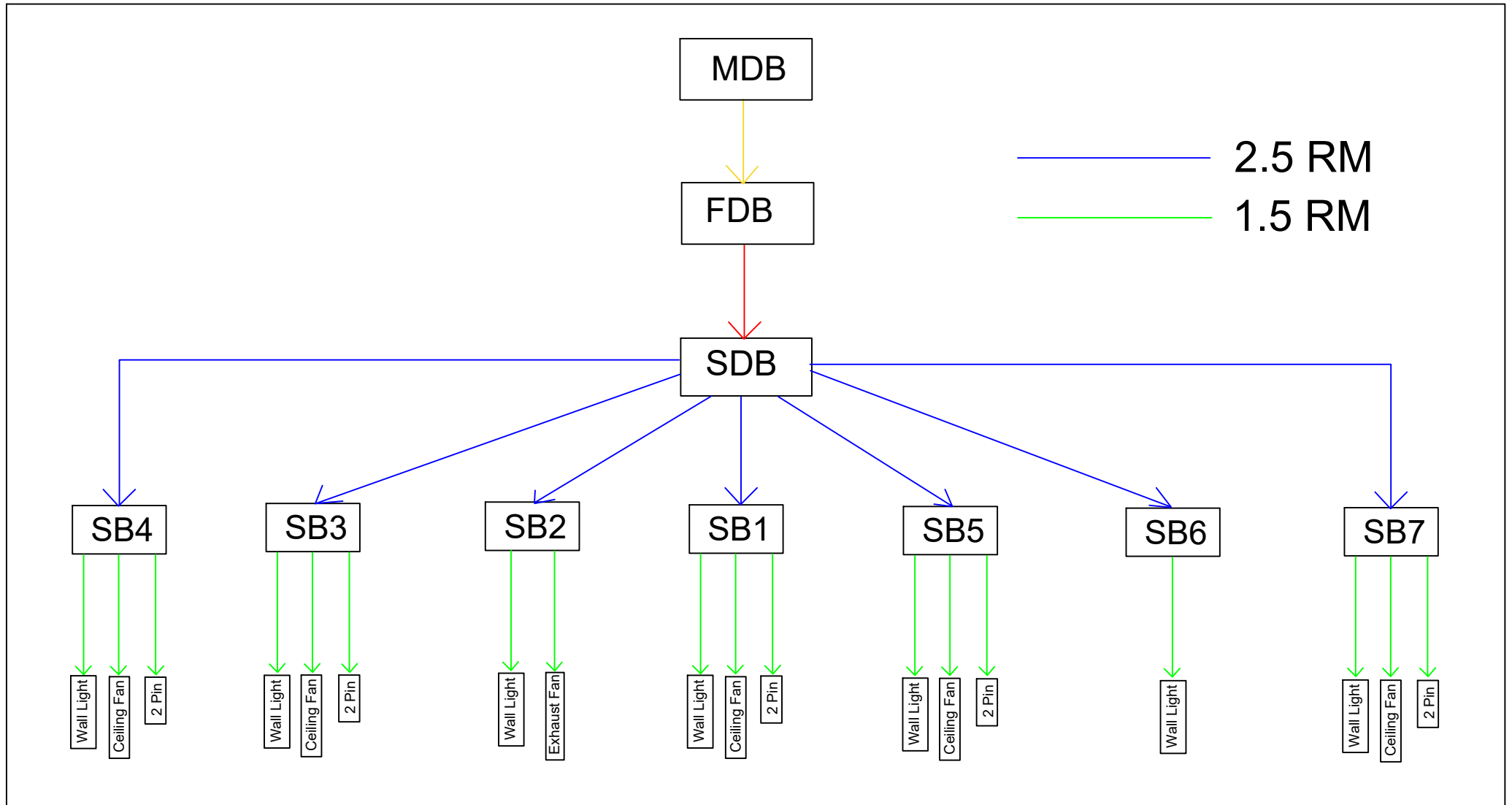
3 Pin

Washing Machine

SB7
WL16
WL17
WL18
CF
2PIN

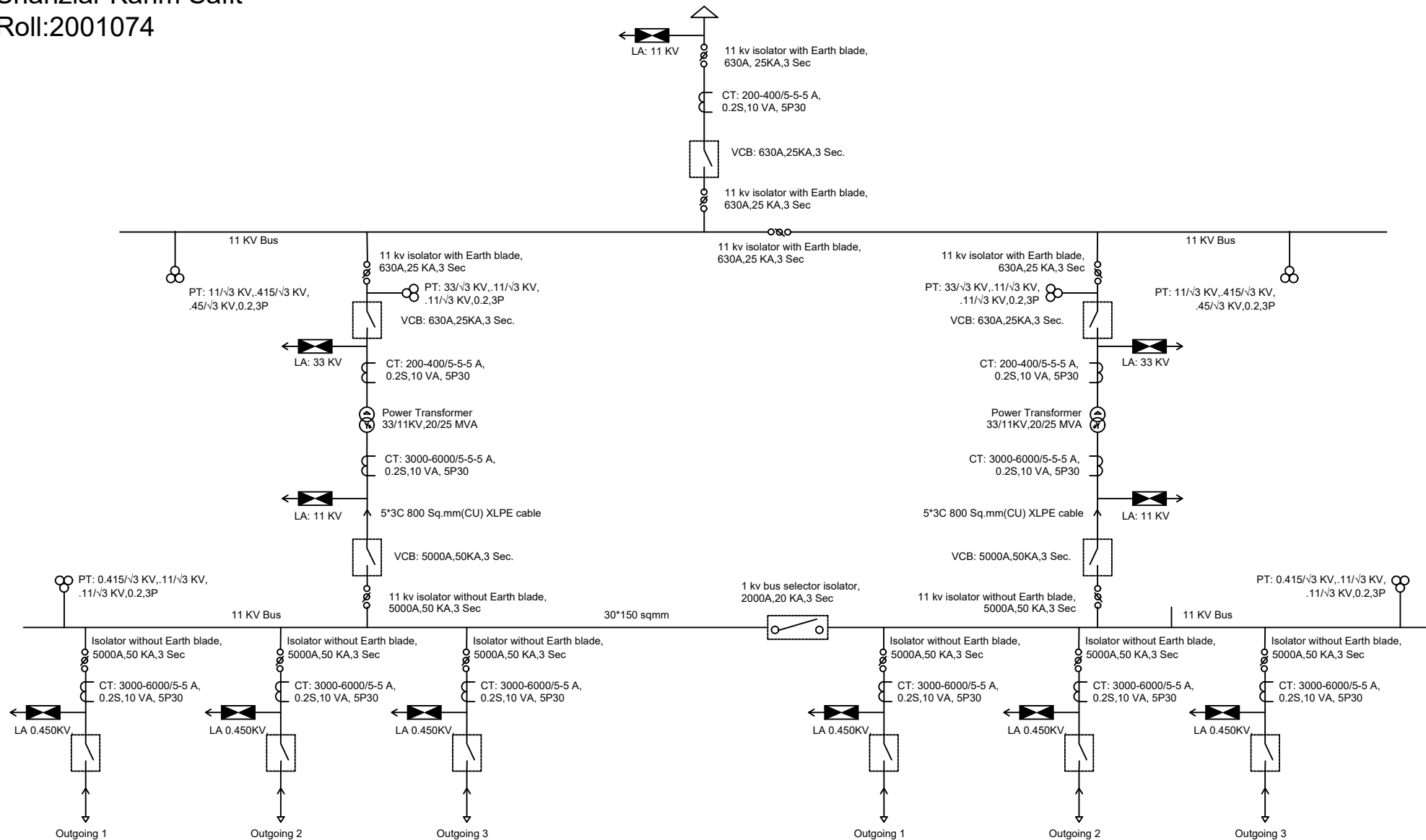
3 Pin

SLD of Sub Circuit of Light Loads of Each Apartments



Shahziar Karim Safit
Roll:2001074

11/0.415 KV Substation



Floor Load Calculation

Light Load Calculation:

Sl. No.	Place	Apartment Load	Qty	Wattage per equipment (W)	Total Wattage per item (W)	Multiplying factor (floor apartmnet number=3)
1	Master Bed room 1, Balcony 1 & Bathroom 1	Wall mounted light	4	60	240	720
		Ceiling fan	1	80	80	240
		2 pin socket	1	300	300	900
2	Bed room 2, Balcony 2 & Bathroom 3	Wall mounted light	4	60	240	720
		Ceiling fan	1	80	80	240
		2 pin socket	1	300	300	900
3	Bed room 3, Balcony 3 & Bathroom 2	Wall mounted light	4	60	240	720
		Ceiling fan	1	80	80	240
		2 pin socket	1	300	300	900
4	Kitchen	Wall mounted light	1	60	60	180
		Exhaust fan	1	100	100	300
5	Dinning Room	Wall mounted light	2	60	120	360
		Ceiling fan	1	80	80	240
		2 pin socket	1	300	300	900
6	Drawing Room	Wall mounted light	2	60	120	360
		Ceiling fan	1	80	80	240
		CCTV	1	30	30	90
7	Floor Lobby	Wall light	11	40	440	1320
TOTAL LIGHT LOAD						10,110 W

LIGHT LOAD SUB-CIRCUIT CALCULATION:

$$\text{Light load total sub circuit} = \frac{10110}{800}$$

$$= 12.64$$

$$\approx 13$$

Power Load Calculation

Sl. No.	Place	Apartment Load	Qty	Wattage per equipment (W)	Total Wattage per item (W)	Multiplying factor(floor apartmnet number=3)
1	Master Bed room 1, Balcony 1 & Bathroom 1	3 pin socket	2	1300	2600	7800
		AC	1	2000	2000	6000
2	Bed room 2, Balcony 2 & Bathroom 3	3 pin socket	2	1300	2600	7800
		AC	1	2000	2000	6000
3	Bed room 3, Balcony 3 & Bathroom 2	3 pin socket	2	1300	2600	7800
4	Kitchen	Oven	1	1500	1500	4500
5	Dinning Room	3 pin socket	1	1300	1300	3900
		Refrigerator	1	1300	1300	3900
6	Store Room	3 pin socket	1	1300	1300	3900
		Washing machine	1	1300	1300	3900
7	Drawing Room	3 pin socket	2	1300	2600	7800
		TV	1	1300	1300	3900
TOTAL POWER LOAD						67,740 W

POWER LOAD SUB-CIRCUIT CALCULATION:

$$\begin{aligned}
 \text{Power load total sub circuit} &= \frac{67740}{2000} \\
 &= 33.87 \\
 &\approx 34
 \end{aligned}$$

TOTAL SUB-CIRCUIT CALCULATION:

Light load total sub circuit = 13

Power load total sub circuit = 34

Total sub circuit = 13+34 =47

Each Apartment Wire Size Calculation (For light load < 800 Watt)

Sl. No.	Location	Equipment Description	Qty	Watt per Equipment (W)	Total Power (W)	Current I = $P/230$ (A)*1.65	Max Current Capacity (A)	Wire Size (mm)	Wire Length (m)	Inew = $I \times 0.93$ (A)	Voltage Drop (V)	Calculated Wire Size (mm)	BNBC Size (mm)	Final Wire Size (mm)
1	Master Bed Room 1 & Balcony	Wall mounted light	4	60	240	1.72	13	3/0.65	20	1.60	1.20	1.0	1.5	1.5
		Ceiling fan	1	80	80	0.574	13	3/0.65	20	0.534	0.41	1.0	1.5	1.5
		2-pin socket	1	300	300	2.152	13	3/0.65	20	2.00	1.50	1.5	1.5	1.5
2	Bed Room 2, Balcony 2 & Bath-3	Wall mounted light	3	60	180	1.29	13	3/0.65	20	1.2	0.9	1.0	1.5	1.5
		Ceiling fan	1	80	80	0.574	13	3/0.65	20	0.534	0.41	1.0	1.5	1.5
		2-pin socket	1	300	300	2.152	13	3/0.65	20	2.00	1.50	1.5	1.5	1.5
3	Bed Room 3, Balcony 3 & Bath-2	Wall mounted light	4	60	240	1.72	13	3/0.65	20	1.60	1.20	1.0	1.5	1.5
		Ceiling fan	1	80	80	0.574	13	3/0.65	20	0.534	0.41	1.0	1.5	1.5
		2-pin socket	1	300	300	2.152	13	3/0.65	20	2.00	1.50	1.5	1.5	1.5
4	Kitchen	Wall mounted light	1	60	60	0.430	13	3/0.65	20	0.400	0.30	1.0	1.5	1.5
		Exhaust fan	1	100	100	0.717	13	3/0.65	20	0.667	0.50	1.0	1.5	1.5
5	Dining Room	Wall mounted light	2	60	120	0.86	13	3/0.66	20	0.80	0.60	1.0	1.5	1.5
		Ceiling fan	1	80	80	0.574	13	3/0.67	20	0.534	0.41	1.0	1.5	1.5
		2-pin socket	1	300	300	2.152	13	3/0.68	20	2.00	1.50	1.5	1.5	1.5
6	Store Room	Wall mounted light	1	60	60	0.430	13	3/0.69	20	0.400	0.30	1.0	1.5	1.5

7	Drawing Room	Wall mounted light	2	60	120	0.86	13	3/0.65	20	0.80	0.60	1.0	1.5	1.5
		Ceiling Fan	1	80	80	0.58	13	3/0.72	20	0.54	0.41	1.0	1.5	1.5
		CCTV	1	30	30	0.21	13	3/0.70	20	0.20	0.15	1.0	1.5	1.5
		2 pin Socket	1	300	300	2.15	13	3/0.80	20	2.00	1.50	1.5	1.5	1.5
8	Floor Lobby	Wall light	11	40	440	3.15	13	3/0.71	20	2.93	2.20	1.0	1.5	1.5

Total Current (Power Load): 25.021 A

Wire Size:

- **Sockets : 1.5 RM**
- **Light/Fan/CCTV: 1 RM**

Each Apartment Wire Size calculation(For Power load > 800 Watt)

Sl. No	Location	Equipment	Qty	Watt / Unit (W)	Total Power (W)	Design Current Id (A)	Cable Ampacity (A)	Wire Size (mm)	Length (m)	Inew = Id×0.93 (A)	V-drop Constant	Voltage Drop (V)	Calculated Wire Size (mm)	BNBC Size (mm)	Final Wire Size (mm)
1	Master Bed Room & Balcony & Bathroom 1	AC	1	2000	2000	14.35	26	7/0.74	20	13.35	0.013	3.73	4.0	4.0	4.0
		3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
		3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
2	Bed Room 2 & Balcony 2	AC	1	2000	2000	14.35	26	7/0.74	20	13.35	0.013	3.73	4.0	4.0	4.0
		3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
		3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
3	Bed Room 3	3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
		3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
4	Kitchen	Oven	1	1500	1500	10.76	20	7/0.67	20	10.01	0.021	4.52	2.5	2.5	2.5
5	Dining Room Dining Room	Refrigerator	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
		3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
6	Store Room	Washing Machine	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5

		3-pin socket		1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
7	Drawing Room	TV	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5
		3-pin socket	1	1300	1300	9.33	20	7/0.67	20	8.68	0.021	3.92	2.5	2.5	2.5

Total Current (Power Load): 151.42 A

Wire Size:

- **ACs : 4.0 RM**
- **Socket/Refrigerator/Washing Machine/Oven : 2.5 RM**

MCB Size Calculation (Light Load)

Sl. No.	Place	Points	Description	Qty	Watt per Equipment (W)	Sub-Circuit Wattle (W)	Current (A) ($1.65 \times P/230$)	Allowable Current (A) (+25%)	MCB Type
1	Master Bed Room-1 & Balcony	6	Wall mounted light	4	60	620	4.45	5.56	1PC6
			Ceiling fan	1	80				
			2-pin socket	1	300				
2	Bed Room-2, Balcony-2 & Bath-3	5	Wall mounted light	3	60	560	4.02	5.03	1PC6
			Ceiling fan	1	80				
			2-pin socket	1	300				
3	Bed Room-3, Balcony-3 & Bath-2	6	Wall mounted light	4	60	620	4.45	5.56	1PC6
			Ceiling fan	1	80				
			2-pin socket	1	300				
4	Kitchen	2	Wall mounted light	1	60	160	1.15	1.44	1PC6
			Exhaust fan	1	100				
5	Dining Room	4	Wall mounted light	2	60	500	3.58	4.48	1PC6
			Ceiling fan	1	80				
			2-pin socket	1	300				
6	Store Room	1	Wall mounted light	1	60	60	0.43	0.54	1PC6
7	Drawing Room	5	Wall mounted light	2	60	590	4.22	5.28	1PC6
			Ceiling fan	1	80				
			CCTV	1	30				
			2-pin socket	1	300				
8	Floor Lobby	11	Wall light	11	40	440	3.15	3.94	1PC6

MCB Size Calculation (For Power Load > 800 W)

Sl. No.	Location	Equipment	Total Power (W)	Design Current Id (A)	Recommended MCB
1	Master Bed Room & Balcony & Bath-1	AC	2000	14.35	DPC25 A
		3-pin socket	1300	9.33	1PC16 A
		3-pin socket	1300	9.33	1PC16 A
2	Bed Room-2 & Balcony-2	AC	2000	14.35	DPC25 A
		3-pin socket	1300	9.33	1PC16 A
		3-pin socket	1300	9.33	1PC16 A
3	Bed Room-3	3-pin socket	1300	9.33	1PC16 A
		3-pin socket	1300	9.33	1PC16 A
4	Kitchen	Oven	1500	10.76	1PC20 A
5	Dining Room	Refrigerator	1300	9.33	1PC16 A
		3-pin socket	1300	9.33	1PC16 A
6	Store Room	Washing Machine	1300	9.33	1PC16 A
7	Drawing Room	3-pin socket	1300	9.33	1PC16 A
		TV	1300	9.33	1PC16 A
		3-pin socket	1300	9.33	1PC16 A

Based on design current, MCB selection is as follows:

For Light Load:

6 A SP MCB (1P C-Curve) is selected for all light load sub-circuits.

For Power Load:

- **AC units:** 25 A DP C-Curve MCB
- **3-Pin sockets / Refrigerator / Washing machine / TV:** 16 A SP C-Curve MCB
- **Oven:** 20 A SP C-Curve MCB

Ground Floor Connected Load Calculation

Sl. No.	Location	Qty	Watt / Unit (W)	Total Power (W)
1	Parking wall lights	16	40	640
2	Guard room light	1	60	60
	Guard room fan	1	80	80
	Guard room 2-pin socket	1	300	300
3	Toilet light	1	60	60
4	Guest room light	1	60	60
	Guest room fan	1	80	80
	Guest room 2-pin socket	1	300	300
	Total Connected Load			1,580 W

Ground Floor Wire Size Calculation

Sl. No.	Location	Equipment Description	Qty	Watt / Unit (W)	Total Power (W)	Design Current Id (A)	Cable Ampacity (A)	Wire Size (rm)	Length (m)	Inew = Id × 0.93 (A)	V-drop Constant	Voltage Drop (V)	Calculated Wire Size (rm)	BN BC Size (rm)	Final Wire Size (rm)
1	Parking Area (GF)	Wall light	16	40	640	4.59	13	3/0.65	20	4.27	0.035	2.99	1.0	1.5	1.5
2	Guard Room	Light	1	60	60	0.43	13	3/0.65	20	0.4	0.035	0.28	1.0	1.5	1.5
		Fan	1	80	80	0.57	13	3/0.65	20	0.53	0.035	0.37	1.0	1.5	1.5
		2-pin socket	1	300	300	2.15	13	3/0.65	20	2.0	0.035	1.4	1.0	1.5	1.5
3	Toilet	Light	1	60	60	0.43	13	3/0.65	20	0.4	0.035	0.28	1.0	1.5	1.5
4	Guest Room	Light	1	60	60	0.43	13	3/0.65	20	0.4	0.035	0.28	1.0	1.5	1.5
		Fan	1	80	80	0.57	13	3/0.65	20	0.53	0.035	0.37	1.0	1.5	1.5
		2-pin socket	1	300	300	2.15	13	3/0.65	20	2.0	0.035	1.4	1.0	1.5	1.5

Ground Floor MCB Selection

Sl. No.	Location	Load Description	Total Load (W)	Design Current Id (A)	Allowable Current (A)	Selected MCB
1	Parking Area (GF)	16 × Wall lights	640	4.59	5.74	
2	Guard Room	Light + Fan + 2-pin socket	440	3.16	3.95	
3	Toilet	Light	60	0.43	0.54	
4	Guest Room	Light + Fan + 2-pin socket	440	3.16	3.95	
	Total (Ground Floor)	Lighting + Fan + 2-pin sockets	1580	11.33	14.16	1PC16 A

Top Floor Connected Load Calculation

Item	Qty	Watt / Unit (W)	Total Power (W)
Roof lights	9	60	540 W
Total Load			540 W

Top Floor Wire & MCB Size Calculation

1. Wire Size Calculation (Top Floor Roof Lighting)

Sl. No.	Location	Equipment Description	Qty	Watt / Unit (W)	Total Power (W)	Design Current Id (A)	Cable Ampacity (A)	Wire Size (rm)	Length (m)	Inew = Id × 0.93 (A)	V-drop Constant	Voltage Drop (V)	Calculated Wire Size (rm)	BN BC Size (rm)	Final Wire Size (rm)
1	Top Floor (Roof)	Roof light	9	60	540	3.87	13	3/0.65	25	3.6	0.035	3.15	1.0	1.5	1.5

2. MCB Size Calculation (Top Floor)

Sl. No.	Location	Load Description	Total Load (W)	Design Current (A)	Allowable Current (A)	Selected MCB	Remarks
1	Top Floor (Roof)	9 × Roof lights	540	3.87	4.84	1PC6 A	Lighting circuit

11/0.415 kV SUBSTATION DESIGN CALCULATION

1. Load Calculation

Item	Calculation	Value
Connected Load	—	1000 kW
Power Factor	—	0.85
Apparent Load	$1000 / 0.85$	1176.5 kVA
Future Allowance (15%)	1176.5×0.15	176.5 kVA
Maximum Demand	—	1353 kVA

2. Transformer Capacity Selection

Item	Calculation	Value
Maximum Demand	—	1353 kVA
Impedance Allowance (6%)	1353×0.06	81.18 kVA
Required Capacity	$1353 + 81.18$	1434.18 kVA
Selected Transformer	Standard rating	2000 kVA

3. Transformer Rated Current

HT Side (11 kV)

Item	Calculation	Value
Rated Power	—	2000 kVA
Rated Current	$2000 / (\sqrt{3} \times 11)$	105 A

LT Side (0.415 kV)

Item	Calculation	Value
Rated Power	—	2000 kVA
Rated Current	$2000 / (\sqrt{3} \times 0.415)$	2784 A

4. Short Circuit Level

Side	Short Circuit Level	Selected Breaker
HT (11 kV)	25 kA	11 kV, 630 A, 25 kA VCB
LT (415 V)	50 kA	415 V, 3200 A, 50 kA ACB

5. CT Selection (25% Safety Factor Applied)

HT Side CT

Item	Calculation	Value
Rated Current	—	105 A
Safety Factor	25%	—
Design Current	105×1.25	131 A
CT Ratio Selected	—	150 / 5 A

LT Side CT

Item	Calculation	Value
Rated Current	—	2784 A
Safety Factor	25%	—
Design Current	2784×1.25	
	—	

6. Final Equipment Selection

Equipment	Rating
Power Transformer	2000 kVA, 11/0.415 kV
HT Breaker	11 kV, 630 A, 25 kA
LT Breaker	415 V, 3200 A, 50 kA
HT CT	150 / 5 A
LT CT	3500 / 5 A

BILL OF MATERIALS (BOM) – ELECTRICAL WORKS (PER APARTMENT)**A. Lighting & Socket Circuits**

Sl No	Item Description	Location	Quantity	Unit Rate (BDT)	Total Cost (BDT)
1	LED Wall Light	All Rooms	18	800	14400
2	LED Tube Light	Lobby	11	1000	11000
3	Ceiling Fan 80W	Bedrooms/Drawing	5	4000	20000
4	Exhaust Fan 100W	Kitchen	1	2000	2000
5	2-Pin Socket	All Rooms	5	400	2000
6	Switch (6A)	All Rooms	68	250	17000
7	PVC Cu Wire 1.5 sq mm	Lighting Circuits	600	70/m	42000
8	PVC Cu Wire 2.5 sq mm	Socket Circuits	300	120/m	36000
9	PVC Conduit & Accessories	Lighting Circuits	Lot	6000	6000

B. Power Load Circuits (>800 W)

Sl No	Item Description	Location	Quantity	Unit Rate (BDT)	Total Cost (BDT)
1	3-Pin Socket (1300W)	Apartment	9	1200	10800
2	Air Conditioner	Bedrooms	2	7000	14000
3	Refrigerator	Kitchen	1	20000	20000
4	Oven	Kitchen	1	5000	5000
5	Washing Machine	Store Room	1	8000	8000
6	TV	Drawing Room	1	45000	45000

7	PVC Cu Wire 2.5 sq mm	Power Circuits	900	120/m	108000
8	PVC Cu Wire 4.0 sq mm	AC Circuits	360	180/m	64800
9	PVC Conduit & Accessories	Power Circuits	Lot	6000/lot	6000

C. Protection & Distribution Equipment

Sl No	Item Description	Location	Quantity	Unit	Total Cost (BDT)
1	MCB 6A SP	Lighting Circuits	6	700	4200
2	MCB 16A SP	Socket Circuits	6	900	5400
3	MCB 20A SP	Power Circuits	6	1100	6600
4	MCB 32A DP	AC Circuits	3	1500	4500
5	RCCB 40A, 30mA	DB	1	5000	5000
6	Distribution Board (SPN)	Apartment	1	6500	6500
7	Earthing Set (Pipe Type)	Apartment	1	10000	10000

Cost Summary :

Catagory	Total Cost
Lighting & Socket Circuits	150400
Power Load Circuits	281600
Protection & Distribution Equipment	42200
Total	474200
Grand Total (3 Apartments)	1422600

GUARD ROOM, GUEST ROOM, STAIR, PARKING AREA, COMMON AREA

Sl. No.	Item Description	Location	Quantity	Unit Rate (BDT)	Total Cost (BDT)
1	Tube Light	Parking Area	10	1000	10000
2	Wall Light	Guard Room	1	800	800
3	Wall Light	Guest Room	1	800	800
4	Ceiling Fan	Guard Room	1	4000	4000
5	Ceiling Fan	Guest Room	1	4000	4000
6	Ceiling Light	Guard Room	1	1,200	1,200
7	Tube Light	Stair	1	1000	1000
8	Tube Light	Common Area	6	1000	6000
9	Wall Light	Washroom	1	800	800
10	Wiring 1.5 sq.mm Cu (PVC)	All Lighting Points	250 m	70 / m	13,500
11	PVC	All Circuits	Lot	5,000	5,000
12	2 Pin Socket	Guard & Guest Room	2	400	800

Total Estimated Cost: 47900

BILL OF MATERIALS (BOM) – MAIN POWER, TRANSFORMER & SWITCHGEAR

Sl. No	Item Description	Specification / Rating	Quantity	Unit Rate (BDT)	Total Cost (BDT)
1	11/0.415 kV Power Transformer	2000 kVA, ONAN, Dyn11	1	5200000	5200000
2	11 kV Vacuum Circuit Breaker (VCB)	630 A, 25 kA, Indoor	1	1800000	1800000
3	11 kV Isolator with Earth Switch	630 A	2	600000	1200000
4	11 kV Lightning Arrester	10 kA	3	150000	450000
5	CT Set	300/5 A, 15 VA, Class 0.5	3	80000	240000
6	PT Set	11 kV / $\sqrt{3}$ / 110 V	1	250000	250000
7	HT XLPE Cable	11 kV, 3C $\times$ 95 sq mm Cu	150	20000	3000000

8	LT Panel (MDB)	415 V, 2500 A, Form IV	1	2500000	2500000
9	ACB	415 V, 2500 A, 50 kA	1	2200000	2200000
10	Busbar System	Aluminium, 2500 A	1	650000	650000
11	Earthing System (Plate Type)	GI Plate with Accessories	2	8000	16000
12	Main LT XLPE Cable	0.6/1 kV, 4C × 300 sq mm Al	100	3500	350000

Justification of CO & PO:

CO (Course Outcome)

Course Outcome (CO)	Justification
CO1	This assignment demonstrates CO1 by applying fundamental electrical engineering concepts in the design of an electrical service system for a modern apartment building. The work includes load calculation, wire sizing, voltage drop analysis, selection of protective devices, earthing design, and preparation of bill of materials in compliance with BNBC standards, integrating theoretical knowledge with practical engineering applications.

PO (Program Outcome)

Program Outcome (PO)	Justification
PO-a	The assignment satisfies PO-a by identifying and analyzing electrical load requirements, selecting appropriate conductors, protection devices, and distribution systems to solve complex engineering problems related to safe and efficient electrical power distribution in residential buildings.
PO-f	This project fulfills PO-f through strict adherence to BNBC codes and safety regulations, proper earthing system design, voltage drop limitation, and ethical engineering practices, ensuring public safety, reliability, and regulatory compliance.
PO-h	The assignment addresses PO-h by incorporating energy-efficient lighting systems, optimized cable sizing, proper circuit segregation, and cost-effective material selection, contributing to sustainable development, environmental responsibility, and economic efficiency.

